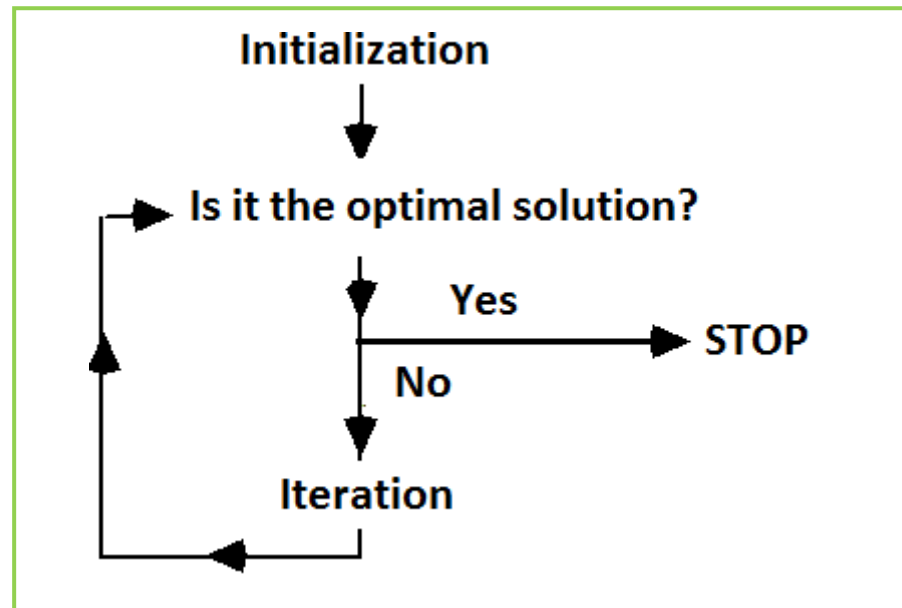


III – The Simplex Method

Introductory concepts

The Simplex method is a generic procedure that enables us to solve linear programming problems.

It is an algebraic method, in which, at each iteration, a system of equations is solved in order to obtain a solution to be tested:



Consider the following definitions:

An **extreme point (EP)** is a point that results from the intersection of any two lines of the constraints.

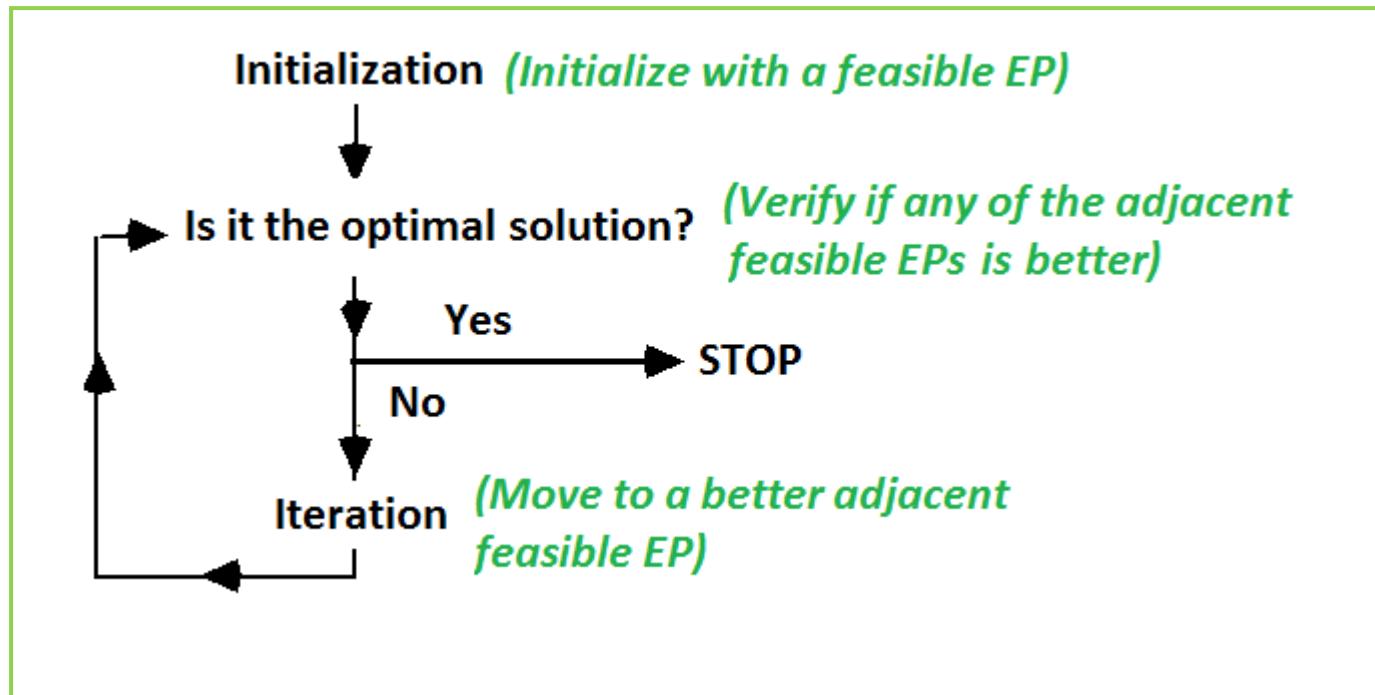
If it belongs to the feasible region is a feasible **EP**; otherwise, it is not feasible.

Two feasible **EPs** are adjacent if they are connected by a single edge in the feasible region.

The essence of the Simplex method is based on geometric concepts, supported on three properties:

- 1 a)** If there is only one optimal solution, then it is a feasible **EP**;
 - b)** If there are multiple optimal solutions then, at least two of them, are adjacent feasible **EPs**;
- 2** There is a finite number of feasible **EPs**;
- 3** If a feasible **EP** has no adjacent feasible **EPs** that are better than it (in terms of the value of the objective function), then, that feasible **EP** is **optimal**.

The Simplex method repeatedly "examines" feasible **EPs** (moving from one **EP** to a better adjacent feasible **EP**), until the current solution has not any better adjacent feasible **EPs**:



We assume that:

- The model is in the *standard* form

$$\text{maximize } z = \mathbf{c}'\mathbf{x}$$

subject to

$$\mathbf{Ax} \leq \mathbf{b}$$

$$\mathbf{x} \geq \mathbf{0}$$

- $b_i \geq 0, \forall i$

Firstly, the inequalities are converted to equalities by the introduction of “slack” variables (one per constraint):

$$\mathbf{Ax} \leq \mathbf{b} \quad \xrightarrow{\text{+ "slacks"}} \quad \mathbf{Ax} = \mathbf{b}$$

The solutions that result from the resolution of the system $\mathbf{Ax}=\mathbf{b}$ are called **basic solutions (BSs)**.

In a problem with n variables and m constraints, a basic solution is composed of:

- m basic variables (**BVs**) – value different of zero
- n non-basic variables (**NBVs**) – value of zero

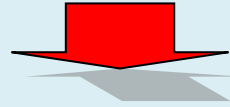
A **basic solution** is **feasible** if all the values of all the variables are ≥ 0 . Otherwise, the solution is not feasible.

Each basic solution corresponds to an extreme point in the graphic:

Feasible EP $\leftrightarrow$ **Feasible BS** (or **FBS**)

Not feasible EP $\leftrightarrow$ **Not feasible BS** (or **NFBS**)

Two **FBS**s are adjacent if they differ in only one **NBV**



It means that moving from one **FBS** to another adjacent **FBS** corresponds to switch one **NBV** with one **BV** (and vice-versa)

Simplex method's algorithm

- a) **Initialization**: How to select the feasible initial **EP**?
- b) **Optimality test**: How to determine that the current feasible **EP** does not have better adjacent feasible **EPs**?
- c) **Iteration**: How to move to a better feasible **EP**?
 - 1. How to select movement direction?
 - 2. When to stop?
 - 3. How to identify the new solution?

a) Initialization:

Any feasible **EP** can be selected as the initial solution:

- before the introduction of the “slacks”: origin;
- after the introduction of the “slacks”: the original variables are the **NBV**s and the “slacks” are the **BV**s;
- null value for the objective function.

b) Optimality test:

Generically, the current feasible **EP** is optimal, if and only if, it does not exist any adjacent feasible **EP** that improves the value of the objective function. That is, if the entry of a **NBV** in the basis does not produce an increase in the objective function value.

c) Iteration:

1. The candidates to enter the basis are the **NBV**s
 - The activity vectors corresponding to the **BV**s are unitary and linearly independent;
 - The activity vectors corresponding to the **NBV**s can be written as a linear combination of them;
 - The **NBV** entering the basis is the one that generates more profit, that is, the one that has associated the highest profitability index:
 - The profitability index corresponding to the entry of one unit of x_j in the basis is given by:
$$c_j - z_j = c_j - \mathbf{c}'\mathbf{B} \mathbf{y}_j$$
 - The profitability index does not specify how many units of the **NBV** x_j should enter the basis.
2. The variable leaving the basis is to first that reaches zero as the value of the variable entering the basis increases.

Example

Let's consider **Example 1** again (page II-1) which mathematical model is:

Determine

x_1 = area to plant of rice

x_2 = area to plant of corn

in order to

maximize $z = 5 x_1 + 2 x_2$

subject to

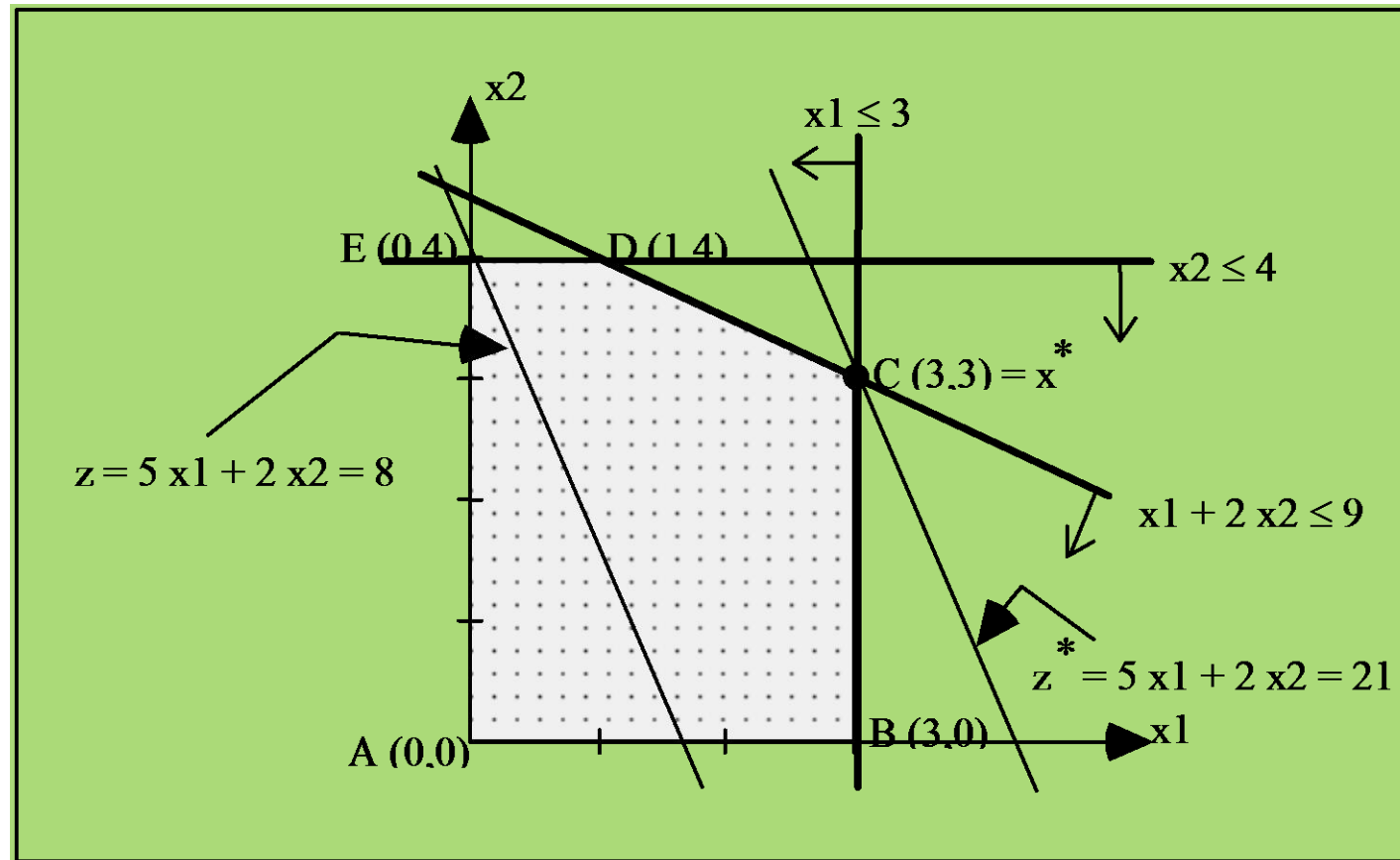
$$x_1 \leq 3$$

$$x_2 \leq 4$$

$$x_1 + 2 x_2 \leq 9$$

$$x_1 \geq 0, x_2 \geq 0$$

Resolution by the graphical method



Resolution by the Simplex method (tabular form)

	c_i	5	2	0	0	0		
x_B	c'_B	x_1	x_2	x_3	x_4	x_5	b	
x_3	0	<u>1</u> *	0	1	0	0	3	(3/1)
x_4	0	0	1	0	1	0	4	
x_5	0	1	2	0	0	1	9	(9/1)
$z_j - c_j$		-5	-2	0	0	0	0	

Extreme point **A**

	c_i	5	2	0	0	0		
x_B	c'_B	x_1	x_2	x_3	x_4	x_5	b	
x_1	5	1	0	1	0	0	3	
x_4	0	0	1	0	1	0	4	(4/1)
x_5	0	0	<u>2</u> *	-1	0	1	6	(6/2)
$z_j - c_j$		0	-2	5	0	0	15	

Extreme point **B**

	c_i	5	2	0	0	0		
x_B	c'_B	x_1	x_2	x_3	x_4	x_5	b	
x_1	5	1	0	1	0	0	3	x_1 optimal
x_4	0	0	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	1	x_4 optimal
x_2	2	0	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	3	x_2 optimal
$z_j - c_j$		0	0	4	0	1	21	z optimal

The table is optimal since there are no negative values in $z_j - c_j$ row.

Optimal solution:

$$\begin{array}{l}
 x_1^* = 3 \\
 x_2^* = 3 \\
 x_3^* = 0 \\
 x_4^* = 1 \\
 x_5^* = 0
 \end{array}
 \left. \vphantom{\begin{array}{l} x_1^* = 3 \\ x_2^* = 3 \\ x_3^* = 0 \\ x_4^* = 1 \\ x_5^* = 0 \end{array}} \right\} \leftrightarrow \text{Extreme point C}$$

Optimal value of the objective function: $z^* = 21$

Simplex method flowchart

